Temperature Estimation with Time Series Analysis from Air Quality Data Set

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***Abstract*— With the expansion of the data size, data mining techniques are gaining more and more importance. Data mining consists of methods such as classification, clustering, time series estimation and association rule. In this study, a time series analysis is carried out in order to make an estimation for the future in accordance with the structure of the data set. Time series are series in which the variables are recorded in chronological order. The data set was created by recording the gas concentrations in the air at a time interval. These data are used to estimate the changes in air quality. Three types of time series analysis training algorithm are used in the study. The results given by the algorithms are close to each other and high performance has been determined. As a result of experimental studies, it is observed that time series analysis is sufficient to estimate air quality.**

Keywords*— *Data Mining, Time Series Analysis, Air Quality

I. Introduction

Today, due to the expansion of the size of data sets, traditional methods are not useful for the analysis of this data. Data mining methods are the process of extracting meaningful information from these large data sets. The large data sets in many scientific areas can be used for the benefit of humanity through these methods. The methods used in data mining are classification, clustering and association rule. Recently, time series analysis in classification algorithms has become the most preferred analysis.

The time series are a data series following a time-dependent order. Time series can make predictions for the future based on present data such as statistics, signal processing, meteorological data, mathematical finance, economics, medicine, chemistry, biology [1]. The best example of the time series is the estimation of the values of the stocks in the stock exchange data [2].

There are many studies on time series in the literature. In the studies carried out, forecasting operations are carried out by using the main components of time series.

Çelik has revealed future accident estimates by applying time-series data of traffic accidents in Turkey. As a result of application exercises, he has foreseen that traffic accidents that occurred the years from 2013 to 2020 in Turkey will increase continuously [3]. Ozden and Ozturk have tested Turkey's energy production subject in two different approaches; time series and Artificial Neural Networks (ANN). As a result of both approaches, it has been found that the time series method has worse estimations than ANN due to lack of data [4]. McAlinn and West investigated how model and estimation options affect time series analysis. The Bayesian perspective defines a new perspective for the density estimation combination using the existing methods in several estimation pools. They have developed a dynamic factor model which is dynamic for time series estimation analysis [5]. Besel and Kayıkçı have worked on the data obtained from the General Directorate of State Meteorology in the Black Sea Region and have made trend analysis for the daily climate and weather data recorded during 32 years in this region [6]. Grange and Carslaw, the data for the air quality with meteorological methods have applied time series analysis using the Random Forest method. Dover, a port city in England by taking advantage of meteorological normalizations, found that the time series were almost exactly estimated [7]. De Vito, Piga and others have installed a multi-sensor device for measuring chemical air quality in urban areas with intense air pollution in Italy. They have collected CO, NO2 on this multi-sensor device, have received total NOx contamination concentration estimates and all results obtained have been processed in MATLAB. According to the results obtained by applying the Levenberg-Marquardt algorithm, they have demonstrated conditions for performing a multivariate calibration [8]. De Vito, Massera, et. al. have associated another study with the Pirelli Laboratories. In his study, ANN have been used to estimate air quality measurements of a multi-sensing device, which is based on seven solid-state sensors, the so-called electronic nose. By using "Neural Time Series Applications" in MATLAB, they have shown that it is possible to develop a neural network that makes an estimate the benzene ratio in the air with the Levenberg-Marquardt algorithm [9].

In the light of these findings, De Vito, Fatturuso, et. al. have proposed two solutions which could be an example of artificial olfactory with the semi-controlled learning approach [10].

As a result, this study has taken advantage of Air Quality Data Set developed by De Vito and his team. The aim of this

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study is to demonstrate "Neural Time Series Applications" in MATLAB and examine the working performance for time series. “Nonlinear Autoregressive with External Input (NARX)”, “Nonlinear Autoregressive (NAR)” and Nonlinear Input-Output Network Models” were used during applications stages in this study. At the same time, the Levenberg - Marquardt algorithm is preferred for both speed and functionality. As a result of experimental studies, it is seen that if a network model suitable for the data set and algorithm are selected, good estimates will be obtained.

In the first section of this paper, general information about the study and literature are given. In the next section, information is given about data set, time series analysis and ANN. In the third section, time series analysis is applied to the data set and the results are shown in the table and figures. The results obtained in the fourth section are evaluated.

II. Material And Method

The Air Quality data set from the UCI Machine Learning Repository site was used. In Italy, a multi-sensor device was installed in an area with high air pollution. The study by De Vito et al. has been a useful data set that has become the subject of many articles [11].

The main objective of the study is to evaluate the classification performance by applying time series analysis to air quality data recorded at a certain time interval. The analysis is carried out using the Neural Net Time Series block, one of the toolboxes in the MATLAB. In the rest of the study, the data set to be used in the application will be mentioned and explanations will be made about the parameters in the time series block. In the result section, the results of the analysis are given to the air quality data set when the time series is applied.

A. Data Set

The data set consists of data recorded continuously with Air Quality Chemical Multisensor Device from March 2004 to February 2005. Various arrangements have been made due to the missing data in the original state of the data set. The missing data tagged with -200 was extracted from the data set. The data set used in the last case consists of 6941 rows and 12 attributes. Chronologically recorded in the columns, CO, tin oxide, benzene, titania, NOx, tungsten oxide, NO2, indium oxide gas concentrations in a cubic meter and the current temperature, relative humidity, and absolute humidity is composed of values. Samples are recorded in rows. Detailed information of the columns is given in Table I.

TABLE I. Properties of The Dataset

Attributes	Max – Min Values	Data Types	Attributes Information
CO	11.9 – 0.1	Real	Hourly concentration ratio CO in mg/m^3
PT08.S1	2040 - 647	Integer	CO targeted Tin Oxide sensor average response
C6H6	63.7 – 0.2	Real	Hourly average of Benzene in microg / m ^ 3
PT08.S2	2214 - 390	Integer	NMHC targeted Titania sensor average response
NOx	1479 - 2	Integer	Hourly average of NOx
PT08.S3	2683 - 322	Integer	NOx targeted Tungsten Oxide sensor average response
NO2	333 – 2	Integer	Hourly concentration ratio NO2 in mg/m^3
PT08.S4	2775 - 551	Integer	NO2 targeted Tungsten Oxide sensor average response
PT08.S5	2523 - 221	Integer	O3 targeted Indium Oxide sensor average response
T	44.6 – (-1.9)	Real	Temperature
RH	88.7 – 9.2	Real	Relative Humidity (%)
AH	2.2 – 0.18	Real	Absolute Humidity

B. Time Series Analysis

In recent years, traditional analysis methods have been replaced by intelligent analysis methods such as fuzzy logic, ANN, support vector machines, and time series. One of the interesting methods of intelligent analysis is time series analysis. Time series are data sets created by recording any size in chronological order. In the time series, at least one parameter must be continuously recorded at certain intervals of time. Time series analysis is used to extract interesting patterns using successively recorded values within a certain interval of time [1]. Time series is preferred because of the simplicity of the structure and easy results in the estimation.

The time series consists of four components. These are;

- Trends,
- Seasonal fluctuations,
- Cyclic component,
- Random (irregular) component.

Trends, indicate the general trend of the series. It is the situation in which the time series is fixed after a long period of increase and decrease. Seasonal fluctuations are the component in which the series shows periodic changes. Rather, it is the factor considered in the analysis of time series related to economics. That is, seasonal fluctuations are periodic changes in which seasonal changes have no effect. Changes that can be shown as error term are not transferred by random component [12].

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The data set must be sufficiently large and reliable to find good results in time series. Time series to provide a positive result means to provide the necessary assumptions. An appropriate time series analysis model should be selected for the data set.

C. *Artificial Neural Networks (ANN)*

Artificial neural networks (ANN) is a technology developed based on the working principle of the human brain. It works with similar logic of biological neural networks. ANN is also called synapses to the nerve cells that mimic the neuron and neurons. As in the human brain, synaptic connections increase and the learning process occurs as they gain experience in ANN. ANN are composed of three layers. The information coming to the input layer is processed in the hidden layer and reaches the output. The general structure of an ANN is shown in Fig. 1.

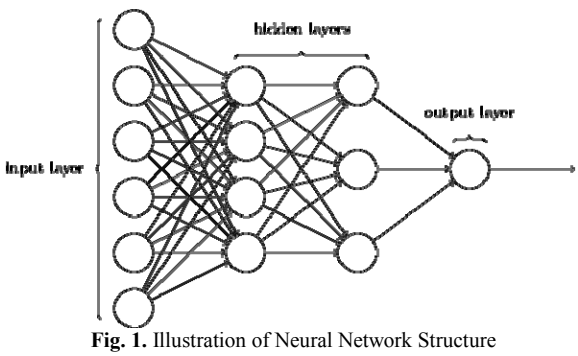

Fig. 1. Illustration of Neural Network Structure

III. Experimental Studies

The data set used in this study presents the measurements of the chemical gases in the air on an hourly basis. Temperature estimation will be made by taking this data into consideration.

Nonlinear Autoregressive with External Input (NARX) network model is used for the analysis this data set. The NARX-type network model uses the previous estimation values and input values together to calculate the next estimation data. The formulation of time series y(t) is given in Figure 2 [14].

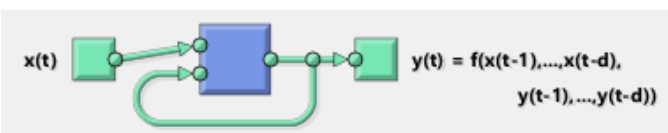

Fig. 2. Nonlinear Autoregressive with External Input

NARX models can be used in problems such as unemployment rates, estimation of economic variables and determining the changing properties of chemicals over time. After pre-process stage, the number of neurons in hidden layer and number of delay is chosen. As shown in Fig. 3, detailed structure of the neural network model is presented.

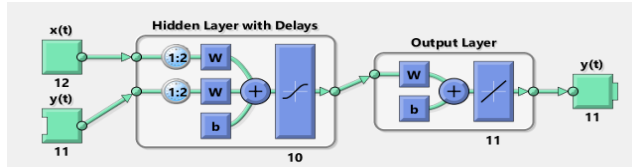

Fig. 3. Neural Network Structure

While NARX makes problem solutions, the data in the hidden layer is transferred to sigmoid function. In the output layer, a two-tiered feedforward applies a linear process. The reason for the efficiency of the NARX network is to return the output y(t) back to the input and use it as input data. Since actual output data is also used when this network is being trained, the resulting network can make a full prediction and provide more accurate input to the feedforward.

In the training stage, three different algorithms are used. These algorithms are given below.

- The Levenberg-Marquardt Algorithm is an algorithm that performs much less time in memory, although it takes up more memory space. It works basically as Quasi-Newton methods. Among these three algorithms, feedback is the fastest performing algorithm. It is suitable for use in algorithm non-large feedforward neural networks.
- Bayesian Regularization Algorithm, which is useful for small and noisy data, takes more time while running this algorithm. It continues training until it finds the best results by minimizing the combinations of square errors and weights. A good generalization can be made with the result obtained because it progresses continuously.
- The Scaled Conjugate Gradient Algorithm determines the weight and tendency values by the conjugate gradient method. As long as there are input, transfer and weight values, they can train each given network by using derivative.

All these experiments are conducted by Matlab software. The Matlab software is a fourth-generation programming language. It is often preferred by users because of the ease of code writing, ready functions and containing various toolboxes [13]. Data set used in this study is performed with the Neural Net Time Series (NNTS) block which is used in ANN applications of the toolboxes offered by MATLAB.

As a result of the implementation of the algorithms, the estimation performance in the data set is evaluated according to the Mean Squared Error (MSE) and Regression (R) values [14]. Since the structure of time series is similar to ANN, firstly, input and output must be determined. Dataset with 12 attributes and 6941 record are used as the input of the model. The tenth attribute in this dataset, namely the temperature, are used as output. Firstly, experiment was conducted with default values of number of hidden layer and number of delay. The default value of the Number of neuron in the hidden layer is equal 10 and the number of delays is equal 2. Secondly, these parameters were set the 20 and 10 respectively. The results of the models in NNTS are shown in Table II.

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TABLE II. EXPERIMENTAL RESULTS

Algorithms		**Results for Neurons =10 and Delay =2**		**Results for Neurons= 20 Delay=10**	
		MSE	R	MSE	R
Levenberg – Marquardt	Training	1.03169e-0	9.93424e-1	7.24942e-0	9.95349e-1
	Validation	1.14371e-0	9.92303e-1	1.15556e-0	9.92902e-1
	Testing	1.28372e-0	9.91935e-1	1.2252e-0	9.92116e-1
Bayesian Regularization	Training	1.02464e-0	9.93420e-1	6.01064e-0	9.96181e-1
	Validation	0.00000e-0	0.00000e-0	0.00000e-0	0.00000e-0
	Testing	1.06055e-0	**9.93234e-1**	1.09512e-0	**9.92688e-1**
Scaled Conjugate Gradient	Training	2.55858e-0	9.83587e-1	1.30475e-0	9.91483e-1
	Validation	2.32137e-0	9.84159e-1	1.43176e-0	9.91179e-1
	Testing	2.21258e-0	9.86356e-1	1.36733e-0	9.91658e-1

As shown in Table II, the best test result obtained with Bayesian Regularization algorithm. Moreover, it is observed that Levenberg-Marquardt algorithm is faster than other algorithms during the experiments. The MSE value given in the data set for the training is very high compared to the validation and the test part. This is because the Levenberg-Marquardt algorithm is more suited to small data sets. It produces too many errors for the data set we have. The graph of this algorithm is given in Figure 4.

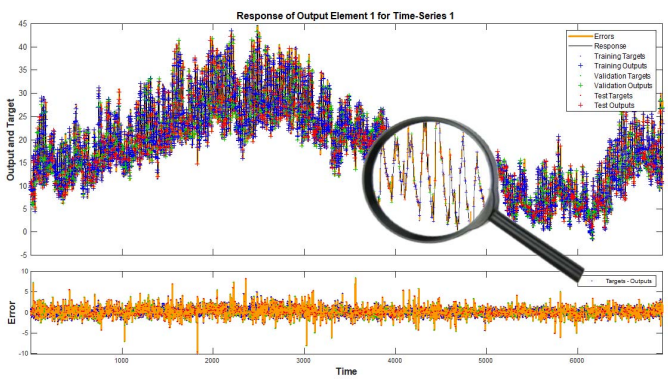

Fig. 4. Response of the Levenberg-Marquardt

The Bayesian Regularization algorithm has slowest running time in these three algorithms. As it does a lot of probabilistic work to minimize error rates, it continues training until it gets the best. As can be seen in Table II, Levenberg-Marquardt algorithm has a lower error rate and higher performance. These results can be seen visually in Figure 5.

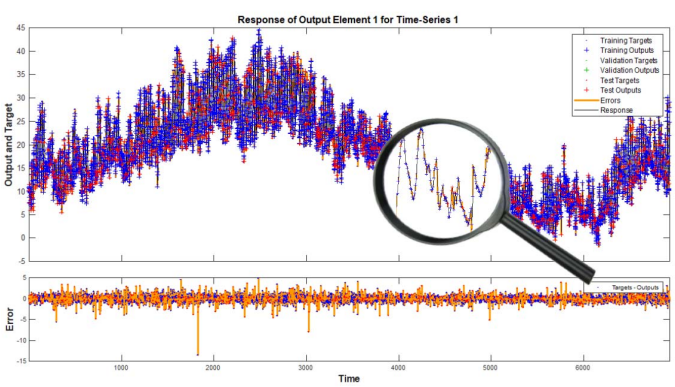

Fig. 5. Response of the Bayesian Regularization

The Scaled Conjugate Gradient algorithm determines the weight and tendency values using the derivative. This algorithm gives the lowest error in the test phase However, validation, test and performance rates are lower than the other two algorithms. Figure 6 also shows the graph of this algorithm.

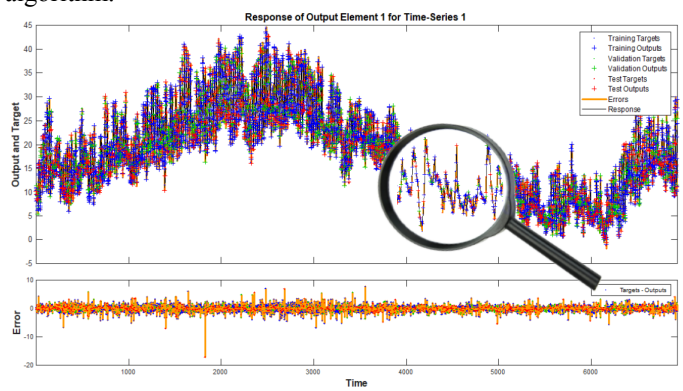

Fig. 6. Response of the Scaled Conjugate Gradient

IV. CONCLUSION

It is always an interesting area to make estimates for the future. Making these estimates scientifically strengthens the accuracy of the results. Time series analysis, which is one of the important methods of data mining, provides forecasted situations based on previously known values. In order for time series algorithms to be efficient, the data set must contain a large number of reliable data.

In this study, Levenberg-Marquardt, Bayesian Regularization, and Scaled Conjugate Gradient algorithms were applied to Air Quality data set within the NARX network model used in time series analysis in MATLAB. The results obtained from these algorithms based on ANN modeling are presented in Table II. According to these results, it is seen that the number of selected neurons and delay has an effect on the performance.

These algorithms were applied to the data set and graphs were obtained. The graphs show the errors and response values for the training, verification and test data. When these three algorithms are analyzed, Bayesian Regularization is the most appropriate algorithm for the air quality data set. It is considered to be the clearest results obtained with this algorithm, since it is more suitable for the size of the data set,

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although it yields later. When the figures and tables are examined, it is seen that the time series is successful in making predictions for the future.

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